SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:**CSA1515

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I Sanjay S/192521058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:
Sanjay S

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
   Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Auto-Scaling and Load Balancing for Seasonal Sales

1. Relevance to Application

An e-commerce company experiences sudden traffic increases during seasonal sales, such as Diwali, Black Friday, or New Year offers. If its servers cannot handle the additional requests, customers may face slow page loading, failed payments, or website downtime.

2. Use of Cloud Concepts

- Auto-scaling: Automatically adds server instances when CPU utilization, request volume, or response time increases and removes excess instances when demand falls.
- Load balancing: Distributes incoming requests across healthy servers to prevent any single server from becoming overloaded.
- Cloud monitoring: Tracks CPU, memory, request rate, latency, errors, and server health.
- Caching: Stores frequently accessed product pages and images to reduce application-server workload.

Customers During Seasonal Sale
Load Balancer
Server 1
Server 2
Server 3 + More
Auto-Scaling + Monitoring

3. Practical Insight

For example, during a major Diwali sale, an online shopping platform may experience a fivefold increase in traffic. Auto-scaling launches additional instances, while the load balancer distributes customer requests among them. Monitoring tools trigger alerts when response time or error rates rise.
This reduces overload-related downtime and improves page loading speed. Capacity limits, database bottlenecks, and payment-service failures must also be managed.

4. Clarity

Conclusion: Auto-scaling adjusts computing capacity to demand, load balancing distributes traffic, and monitoring detects problems early. Together, these concepts improve availability and customer experience while reducing the cost of idle servers outside peak periods.

2. Secure and Cost-Effective Cloud Backup Storage

1. Relevance to Application

An organization needs to store large volumes of backup data securely while ensuring that files can be recovered after accidental deletion, cyberattacks, hardware failure, or natural disasters.

2. Use of Cloud Concepts

- Durability: Cloud object storage is designed to protect stored data against hardware failures and data loss.
- Redundancy: Multiple copies or erasure-coded data are maintained across devices or failure domains, depending on the storage service.
- Encryption: Data is encrypted at rest and in transit using appropriate encryption mechanisms.
- Access control: IAM policies, least privilege, MFA, and separate backup accounts restrict unauthorized access.
- Versioning and immutability: Previous versions and protected backup copies help recover deleted or modified data.
- Lifecycle management: Older backups can be moved to lower-cost archival storage.

3. Practical Insight

For example, a hospital can back up patient records and appointment databases to cloud object storage. If ransomware encrypts its production systems, the hospital can restore clean copies from an isolated, immutable backup.
Compared with traditional storage, cloud backup can reduce upfront hardware purchases, physical maintenance, and data-center space. However, retrieval fees, data transfer charges, retention periods, and recovery time must be considered.

4. Clarity
Conclusion: Cloud backup combines redundancy, encryption, access control, and lifecycle management to protect data at a scalable cost. Regular restoration tests and isolated backup copies are essential for reliable disaster recovery.

3. CDN for Global Mobile Applications
1. Relevance to Application
A mobile application serving users across different countries must deliver videos, images, game assets, and other content with minimal latency and consistent performance.

2. Use of Cloud Concepts
- Content Delivery Network (CDN): Distributes cached content across geographically dispersed edge servers.
- Edge servers: Serve content from locations closer to users instead of retrieving every request from a distant origin server.
- Global load balancing: Routes requests to suitable healthy endpoints.
- Cloud availability: Multiple regions, replicated content, and health-based routing help maintain service during regional failures.

Origin Server / Cloud Storage
Global CDN Network
Asia Edge
Europe Edge
US Edge
Users receive cached content from nearby locations whenever possible.

3. Practical Insight
For example, a global video-streaming application can cache popular video segments at edge locations. A user in India receives content from a nearby edge server rather than repeatedly downloading it from a server in the United States.
This reduces latency, network congestion, and buffering. For real-time multiplayer gaming, CDNs help distribute game downloads and updates, while gameplay traffic usually requires low-latency regional game servers rather than ordinary content caching alone.

4. Clarity
Conclusion: CDNs and edge servers improve global performance by reducing the distance content travels. Regional redundancy and health-based routing further improve availability, while dynamic content must be handled with appropriate caching and security policies.

4. Compliance and Security for a Financial Institution

1. Relevance to Application
A financial institution processes sensitive information such as account details, customer identities, payment records, and transaction histories. It must protect this data and satisfy applicable legal, regulatory, and audit requirements.

2. Use of Cloud Concepts
- Compliance controls: Cloud providers offer security certifications, compliance reports, and services that support applicable standards and regulations. Using a compliant service does not automatically make the customer's entire system compliant.
- Logging and auditing: Record authentication events, administrator actions, data access, and transaction-related activities.
- Monitoring: Detect unusual access, repeated login failures, suspicious transactions, and configuration changes.
- Encryption: Protect sensitive data at rest and in transit.
- Identity and access management: Enforce MFA, least privilege, separation of duties, and periodic access reviews.
- Data sovereignty: Select approved regions, control replication and backups, and verify where data and associated records are processed and stored.

3. Practical Insight
For example, when a customer transfers money through a banking application, the system encrypts communication using TLS, verifies the customer's identity, checks transaction permissions, and records relevant audit events. Monitoring systems can flag unusual transaction patterns for investigation.
The institution should review the applicable financial regulations, contractual requirements, and data-localization rules. Encryption keys and access permissions should also be governed according to organizational policy.

4. Clarity
Conclusion: Cloud providers supply tools for encryption, monitoring, auditing, and compliance support. The financial institution remains responsible for correct configuration, access governance, transaction controls, and ensuring that data stays within legally permitted locations.

5. Containers, Virtual Machines, and Kubernetes
1. Relevance to Application
A development team wants to deploy an application consistently across development, testing, and production environments, including different cloud providers.

2. Use of Cloud Concepts

Feature	Containers	Virtual Machines
Architecture	Share the host OS kernel	Run a separate guest OS
Startup	Usually fast	Usually slower
Resource usage	Generally lower overhead	Generally higher overhead
Isolation	Process-level and kernel-based isolation	Hardware-virtualized isolation
Portability	Package application and dependencies	Package or manage a full VM image
Best use	Microservices and rapid deployment	Different OS requirements and stronger isolation boundaries

Containers are lightweight, but they still require appropriate security controls. Virtual machines may provide a stronger isolation boundary for some workloads.
Kubernetes orchestration automates container scheduling, service discovery, health checks, rolling updates, restart of failed containers, and horizontal scaling.

3. Practical Insight
Consider an online shopping application built using microservices:
Customer Web / Mobile Application
API Gateway / Load Balancer
User Service
Login and profiles
Product Service
Catalog and search
Order Service
Orders and status
Payment Service
Payment processing
Kubernetes + Container Registry + CI/CD
Each service runs in its own container and communicates through defined APIs. During a sale, Kubernetes can increase the number of product-service replicas without scaling every other service equally. The same tested container image can be deployed across compatible Kubernetes environments in different clouds.

4. Clarity
Conclusion: Containers improve deployment speed and resource efficiency, while Kubernetes simplifies scaling, recovery, and service management. Container portability supports multi-cloud deployment, although differences

in networking, storage, identity, and cloud-specific services must still be addressed.